Does a Bankruptcy Model Trained in Calm Times Survive a Recession?

A Temporal Stress Test on U.S. Public Firms

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Corporate-distress models are typically trained and validated on data pooled across time, yet they are deployed forward into economic conditions that may differ sharply from the training period. We ask whether a bankruptcy classifier trained on *calm* pre-crisis years still functions during a *recession*. Using 78,682 company-years of U.S. public firms (1999–2018) with 18 financial variables, we frame imminent-failure prediction—a firm's final reporting year before delisting—and engineer 15 Altman-style financial ratios. Training on 1999–2006 with a company-grouped split to prevent leakage, a random forest attains a held-out ROC-AUC of 0.755, well ahead of logistic regression (0.646) and a multilayer perceptron (0.625); the distress boundary is strongly nonlinear. Stress-testing the frozen calm-era models on the 2008–2009 recession, the random-forest point estimate rises to 0.798 AUC, while the linear and neural estimates slip. Analytic 95% confidence intervals overlap across periods, so the supported conclusion is preservation rather than statistically established improvement. We also show that a naive per-year AUC curve suggests a spurious "cliff" at 2007 that is an evaluation artifact of training-set overlap, underscoring the importance of leakage-free temporal validation.

CCS Concepts: • **Applied computing: Economics** • **Computing methodologies: Supervised learning by classification** • **Computing methodologies: Machine learning**

Additional Key Words and Phrases: bankruptcy prediction, distribution shift, temporal validation, financial ratios, class imbalance

1 INTRODUCTION

Models that flag firms at risk of failure are central to credit and investment decisions. Their operational value depends on *temporal* robustness: a model fit on historical data must keep working as macroeconomic conditions evolve. The

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2008 financial crisis is a natural stress test—a sharp regime change between the calm mid-2000s and the recession. We pose a direct question: *does a distress model trained on calm years still rank failing firms during the recession, and which model family is most robust?*

1.1 Contributions

(1) An imminent-failure formulation of the widely used U.S. bankruptcy panel, in which only a firm's final reporting year counts as positive, paired with company-grouped validation so that no issuer is ever split across a training and an evaluation partition.
(2) A calm-era baseline across three model families, in which a random forest clearly outperforms linear and neural alternatives, locating the difficulty of the problem in the nonlinearity of the distress boundary rather than in the feature set.
(3) A temporal stress test that freezes the calm-era models and applies them to the 2008–2009 recession, finding that the random-forest point estimate is preserved and rises modestly, while the linear and neural estimates slip.
(4) A cautionary analysis of a per-year evaluation artifact that can masquerade as a crisis-induced collapse, and which disappears once training-set overlap is removed.

2 RELATED WORK

Distress prediction dates to Beaver's univariate ratio analysis [2], Altman's Z-score [1], and Ohlson's logit model [14], whose ratio families we adopt. Hazard formulations that respect the panel structure of firm data followed [4, 17]. Machine-learning approaches, including random forests [3], are now standard, and tree ensembles remain strong on tabular problems of this shape [7]. The robustness of predictive models under temporal or covariate shift is the subject of the dataset-shift literature [15], and the specific failure mode we document in Section 5.4 belongs to the literature on leakage in applied machine learning [9, 10]. We combine these threads into an explicit calm-to-recession transfer evaluation on a large panel of U.S. firms.

3 DATA

We use a public dataset of 78,682 company-years for U.S. public companies, 1999–2018, each with 18 accounting variables (assets, liabilities, income-statement items, market value) and an `alive`/`failed` status. Failure is rare, so plain accuracy is uninformative and we evaluate with ranking and precision–recall metrics (Figure 1). Distressed firms differ systematically from healthy ones—lower profitability and higher leverage—which is the signal our models exploit.

3.1 Target definition

A failed firm appears in the panel for several years, every one of them stamped `failed`. Labelling all of those years as bankruptcies would be misleading, because a firm five years from delisting usually looks healthy. We therefore label a company-year positive if and only if it is that firm's *final* reporting year,

$$y_{it} = \mathbb{1}\left[\text{status}_i = \texttt{failed} \;\wedge\; t = \max\{t' : (i, t') \in \mathcal{D}\}\right], \tag{1}$$

and we drop the earlier years of failed firms so that the negative class is clean. This yields 609 positives among 74,071 company-years, a base rate of 0.82%. The label is a proxy: a firm's last filing is not always the year it legally enters bankruptcy.

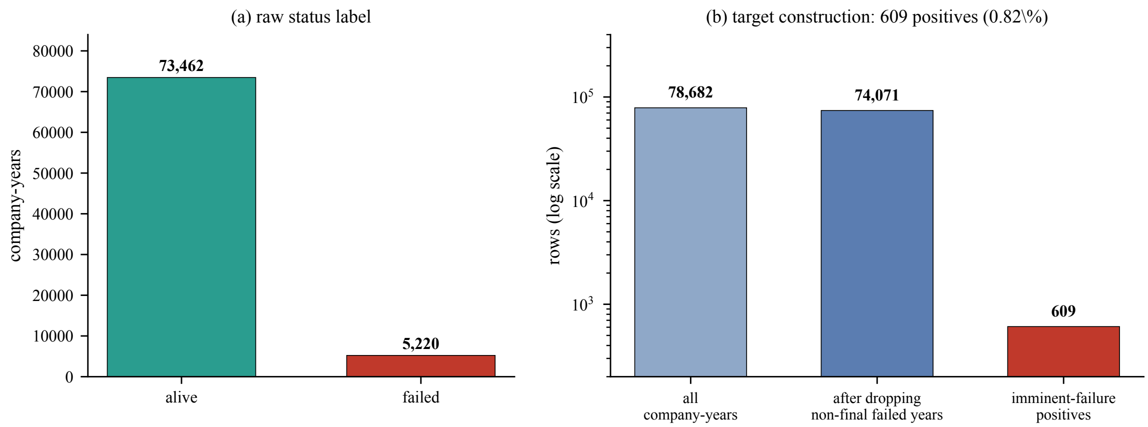

Fig. 1. (a) Raw status labels: 6.6% of company-years belong to firms that eventually failed. (b) The imminent-failure target is far rarer still. Dropping the non-final years of failed firms leaves 74,071 rows, of which only 609 (0.82%) are positive.

3.2 Features

Raw dollar magnitudes are size-confounded—a $1M loss is fatal for a small firm and a rounding error for a large one. From the 18 raw variables we therefore build 15 financial ratios [1, 14]: return on assets, EBIT/assets, EBITDA/assets, working-capital/assets, retained-earnings/assets, sales/assets, leverage, long-term-debt/assets, market-value/liabilities, current ratio, gross and net margins, inventory/assets, receivables/assets, and $\log(1 + \text{total assets})$ as a size term. Each is winsorized at the training-set 1st and 99th percentiles, median-imputed, and standardized, with all three statistics estimated on the training rows only:

$$\tilde{x}_j = \frac{\mathrm{clip}\left(x_j;\, q^{\mathrm{tr}}_{0.01,j},\, q^{\mathrm{tr}}_{0.99,j}\right) - \mu^{\mathrm{tr}}_j}{\sigma^{\mathrm{tr}}_j}. \tag{2}$$

4 METHODS

4.1 Models and metrics

Logistic regression predicts $\hat{p}(x) = \sigma(w^\top \tilde{x} + b)$; the random forest [3] averages 400 decision trees with a minimum of three samples per leaf; the MLP applies two ReLU hidden layers (64 and 32 units) with dropout 0.3 and a sigmoid output. It uses Adam [11] at learning rate 10^{-3}, batch size 256, at most 120 epochs, and early stopping with patience 12 on a 15% validation split. This was one prespecified compact architecture, not the result of an architecture search. With only about 150 positive training rows, its lower AUC is consistent with data scarcity and limited tuning rather than evidence that neural models are intrinsically unsuitable. Given the extreme imbalance, all models use class weighting; the MLP minimizes a weighted binary cross-entropy

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\left[\beta_1 y_i \log \hat{y}_i + \beta_0 (1 - y_i)\log(1 - \hat{y}_i)\right], \qquad \beta_c = \frac{N}{2N_c}. \tag{3}$$

We report ROC-AUC, the probability that a failing firm is ranked above a surviving one [6],

$$\mathrm{AUC} = \Pr\left(\hat{p}(x^+) > \hat{p}(x^-)\right), \tag{4}$$

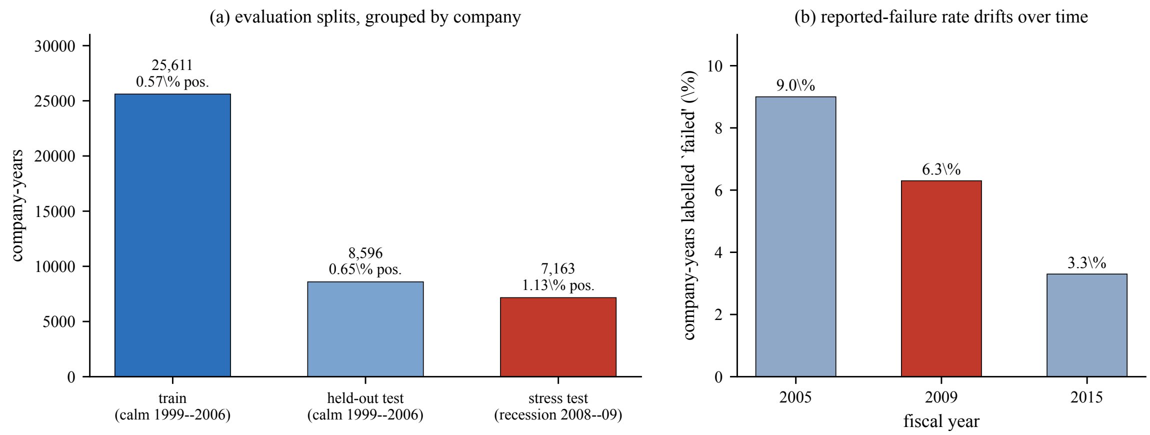

Fig. 2. (a) The three evaluation partitions, with sizes and positive rates. The recession window carries roughly twice the imminent-failure rate of the calm training data (1.13% versus 0.57%). (b) The share of company-years labelled `failed` drifts substantially over the panel.

and PR-AUC (average precision), whose no-skill baseline equals the positive rate. With a base rate near 0.6%, PR-AUC is the more demanding summary and the more honest one for a rare-event problem [5, 16]. We accompany AUC point estimates with Hanley–McNeil large-sample 95% confidence intervals computed from the observed positive and negative counts [8].

4.2 Validation protocol

The calm era (1999–2006) is split *by company* using a grouped shuffle split, so that a firm never appears in both training and calm-test partitions. This matters because each firm contributes several highly correlated yearly rows; a random row split would let the model recognise a company rather than learn a warning sign. The recession window (2008–2009) is held out entirely for the stress test. The NBER dates the recession's onset to December 2007 [13]; fiscal-year 2007 therefore mixes mostly pre-recession activity with the beginning of the contraction. We exclude it rather than assign this transition year to either regime. Figure 2 summarises the resulting partitions.

We define the calm-to-recession *change* as

$$\Delta = \mathrm{AUC}_{\mathrm{recession}} - \mathrm{AUC}_{\mathrm{calm}}, \tag{5}$$

where both terms are computed on firms excluded from training. A non-negative Δ means the model survives the regime shift.

5 EXPERIMENTS AND RESULTS

5.1 Calm-era baseline

Table 1 reports held-out calm-era performance. The random forest (0.755 AUC) clearly beats logistic regression (0.646) and the MLP (0.625): the mapping from financial ratios to imminent failure is strongly nonlinear and interaction-heavy—low profitability *and* high leverage *and* weak liquidity—which the tree ensemble captures and the linear and shallow models do not. PR-AUC is low throughout, an honest consequence of the $\sim$0.6% base rate; even so, the random forest's 0.020 is roughly 3× the no-skill baseline.

Table 1. Held-out calm-era (1999–2006) performance, company-grouped split. The PR-AUC no-skill baseline is the positive rate, 0.0065.

Model	ROC-AUC	PR-AUC
Logistic Regression	0.646	0.010
Random Forest	**0.755**	**0.020**
MLP	0.625	0.012

Table 2. Calm-era versus 2008–2009 recession, both on firms excluded from training. Δ is defined in Eq. (5).

Model	Calm AUC	Recession AUC	Δ
Logistic Regression	0.646	0.615	−0.031
Random Forest	0.755	**0.798**	**+0.042**
MLP	0.625	0.575	−0.050

The corresponding AUC 95% intervals are $[0.567, 0.725]$ for logistic regression, $[0.681, 0.829]$ for the random forest, and $[0.546, 0.704]$ for the MLP. The intervals reinforce the forest's advantage over the two lower point estimates while showing the uncertainty created by only 56 positive calm-test rows.

5.2 The 2008–2009 stress test

Table 2 and Figure 3 report the frozen calm-era models applied to the recession. The random forest does not degrade; its AUC *rises* to 0.798 ($\Delta = +0.042$). The linear and neural models slip modestly ($\Delta = -0.031$ and -0.050). The recession estimate is consistent with financial distress remaining at least as separable as in calm times; it does not establish that crisis failures are universally easier to identify. By ranking, the model survives the stress test.

Recession AUC 95% intervals are $[0.549, 0.681]$ (logistic), $[0.740, 0.856]$ (random forest), and $[0.510, 0.640]$ (MLP). The random forest's calm and recession intervals overlap, so its +0.042 change is a descriptive point-estimate difference, not a statistically resolved improvement.

Two points qualify the headline. First, the recession's higher base rate (1.13% versus 0.57%) raises the attainable precision independently of model quality, so part of the PR-AUC improvement is mechanical rather than evidence of better discrimination; this is why we make ROC-AUC the primary endpoint. Second, the ranking that survives is not a calibrated probability. A threshold tuned on calm-era scores will fire at a different rate when the underlying failure rate roughly doubles, so operating points must be re-calibrated even though the ordering is preserved [18].

5.3 Which financial signals drive the forest?

After completing evaluation, we refit the prespecified forest to all calm-era rows for descriptive interpretation. Table 3 reports its five largest impurity-based importances. Market value relative to liabilities and leverage dominate, followed by liquidity, working capital, and profitability. This agrees with the classic distress literature but is associative, not causal, and impurity importance can favour variables offering more candidate split points.

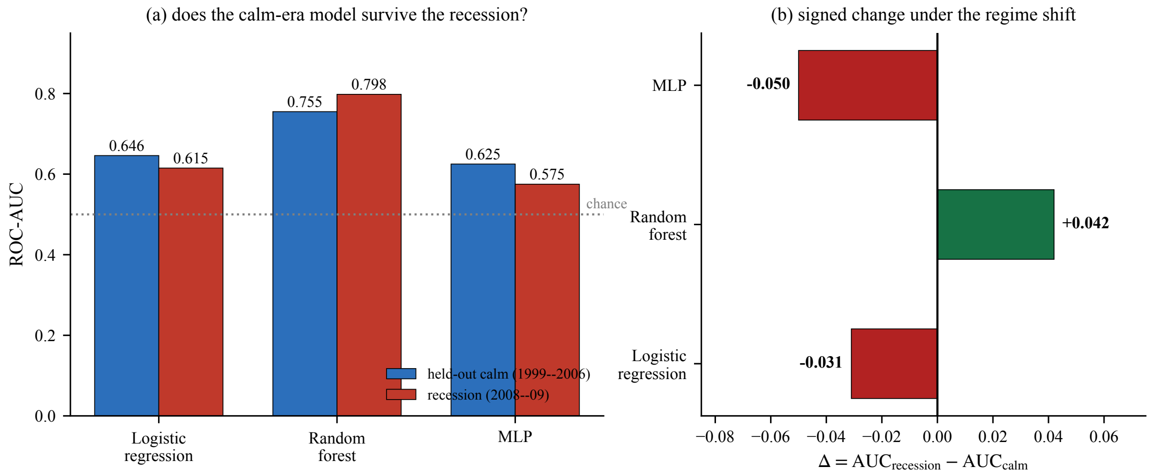

Fig. 3. (a) Calm versus recession ROC-AUC per model. Taller red than blue means the model holds up. (b) The signed change of Eq. (5): only the random forest is positive, and it is the only model that was strong to begin with.

Table 3. Largest random-forest feature importances after a descriptive refit to all calm-era rows. Importances sum to one across all 15 ratios.

Financial signal	Importance
Market value / liabilities	0.142
Leverage (liabilities / assets)	0.119
Current ratio	0.075
Working capital / assets	0.063
Return on assets	0.062

5.4 A per-year artifact that looks like a crisis break

It is natural to diagnose temporal robustness by scoring the calm-trained model on every fiscal year and plotting the resulting AUC curve. Doing so produces an apparent “cliff” after 2006 that invites the reading that the crisis broke the model. It is largely an evaluation artifact.

The mechanism is straightforward. Pre-2007 years still contain many of the exact firms the model trained on, and because a firm’s yearly rows are highly correlated, scoring those years measures partial memorisation rather than prediction. Later years contain genuinely unseen firms. The apparent drop is therefore not the crisis arriving; it is the evaluation finally becoming honest. This is a textbook instance of the leakage patterns catalogued in [9, 10]: the contaminated estimate is not merely optimistic, it induces a qualitatively wrong conclusion about *when* performance changed.

The leakage-free comparison shows no break at all. Figure 4 places the two valid estimates side by side: held-out calm firms at 0.755 and held-out recession firms at 0.798. The correct reading of the same underlying model is that performance rose across the regime change.

The practical rule follows directly: in panel data, temporal validation must exclude a firm’s *entire* history from the test period, not merely the test year.

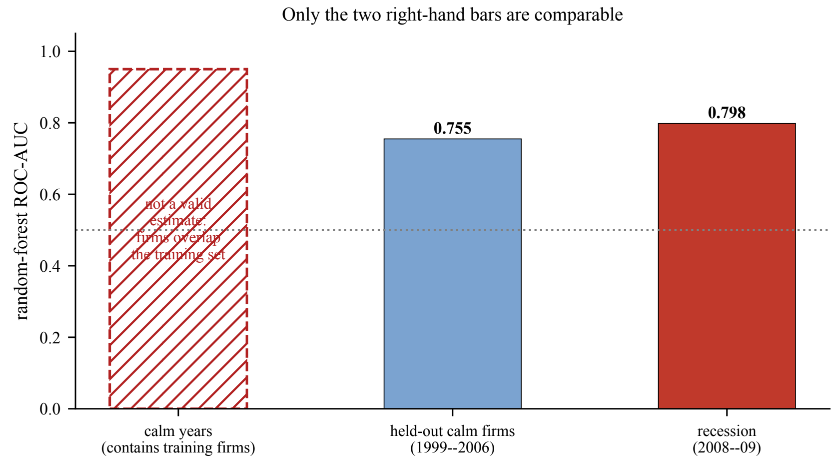

Fig. 4. Only estimates computed on firms excluded from training are comparable. A per-year curve scored on calm years is contaminated by training-firm overlap (left, hatched) and cannot be placed on the same axis as the two valid estimates.

6 DISCUSSION

The headline is reassuring and slightly counterintuitive: a distress model built before the crisis survives the crisis, and the most flexible model is also the most robust. Because financial ratios encode the mechanics of insolvency—eroding profitability, mounting leverage, shrinking liquidity—the calm-era decision surface transfers to the recession, where these signatures are sharper.

That the random forest is simultaneously the most accurate and the most stable by point estimate deserves comment, because flexibility and robustness are often assumed to trade off. Here the distress boundary is genuinely nonlinear, so the linear model is misspecified in both regimes and has no robustness advantage to offer; its smaller absolute loss under shift comes with a much lower absolute score. The MLP, fitting ~ 150 positives, is data-starved rather than over-flexible. The practical caveats concern calibration and measurement, not ranking.

7 LIMITATIONS

Positives are rare, so PR-AUC is low and the recession estimates rest on a few dozen failures; they should be read as directional. The imminent-failure label—the final reporting year—is a proxy for the legal bankruptcy date. Firms recur across the calm and recession windows, so the transfer is across *time* rather than to entirely new firms; this is realistic for re-scoring a portfolio but is a caveat for generalization to unseen issuers. The analytic AUC intervals treat company-years as independent and may be too narrow because surviving firms contribute repeated rows; a company-cluster bootstrap of saved predictions would be preferable. We report one split and seed, so small differences remain uncertain. Finally, survivorship and reporting biases affect which firms appear in the panel at all.

8 MODEL CARD

Table 4 states the intended and prohibited uses of the calm-era random forest in the format of [12]. Every entry restates a decision already made in the sections above; it is collected here because the distinction between a ranking that survives a regime change and an operating point that does not is exactly the distinction a deployment reader needs.

Table 4. Model card [12] for the calm-era random forest.

Item	Statement
Intended use	Research-grade ranking of U.S. public company-years by imminent-failure risk.
Training population	Calm-era company-years, 1999–2006, split by company.
Evaluation population	Held-out calm-era firms and the 2008–2009 recession window; 2007 is excluded as a transition year.
Positive class	A firm's final reporting year before delisting, Eq. (1); base rate 0.82%.
Features	The 15 Altman-style ratios of Equation (2), winsorized, median-imputed and standardized on training rows only.
Primary metric	ROC-AUC. PR-AUC is reported alongside and is the more demanding summary at this base rate.
Decision threshold	None. The model is evaluated purely by ranking; no operating point is fitted or recommended.
Known weak point	Scores are not calibrated probabilities. The recession base rate roughly doubles, so a threshold set on calm-era scores fires at the wrong rate and must be re-calibrated.
Prohibited use	Credit decisions about individual firms, regulatory capital calculations, or any use that reads the score as a probability of bankruptcy.

9 CONCLUSION

On a large panel of U.S. public firms, a random forest trained on calm pre-crisis years predicts imminent bankruptcy at 0.755 ROC-AUC and *survives* the 2008–2009 recession, with a point estimate of 0.798 while simpler models slip. The confidence intervals overlap, supporting preservation but not a resolved improvement. A per-year evaluation artifact can falsely suggest a crisis-induced collapse. We recommend leakage-free temporal validation and threshold re-calibration for deployment across economic regimes.

REPRODUCIBILITY

The analysis notebook fixes all random seeds, defines the target and the company-grouped split explicitly, and estimates every preprocessing statistic inside the training partition. The source panel is licensed and is not redistributed here; the figure script accompanying this manuscript regenerates all figures and tables from the recorded results, and recomputes the per-year analysis when the raw CSV is supplied.

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